

Aprenentatge Automàtic per a Xarxes (ML4Net)

Lecture 10: AI/ML in Wi-Fi Standardization & Mock Test

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1. AI/ML in Wi-Fi standardization

2. Mock Test

The Importance of Standardization

The standardization of communications technologies (e.g., Wi-Fi, Bluetooth, 5G) consists of establishing common rules, names/formats, messages, and mechanisms.

- **Interoperability:** Ensures devices from different manufacturers can communicate seamlessly.
- **Global reach:** Enables widespread use of the technology worldwide.
- **Ensure Quality:** Key to ensuring clearly defined performance goals (e.g., performance, security).
- **Innovation:** Create a common ground to be evolved collaboratively.

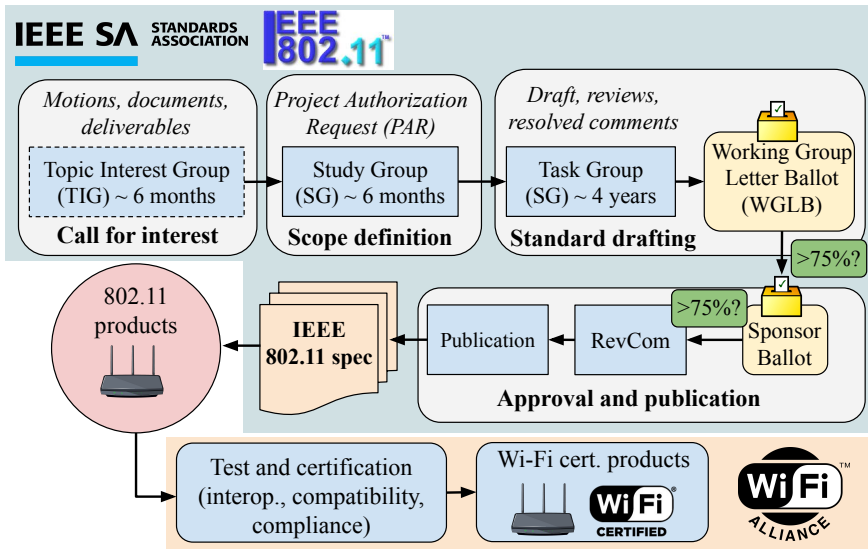
IEEE 802.11 standardization:

- The Institute of Electrical and Electronics Engineers (IEEE) is the primary standardization body for Wi-Fi.
- The **IEEE 802.11 working group** is responsible for developing and maintaining standards for wireless local area networks (WLANs).
- Each major revision is designated by a letter suffix (e.g., 802.11a, 802.11b, 802.11g, 802.11n, 802.11ac, 802.11ax).

The role of the Wi-Fi Alliance (WFA)

- Certifies Wi-Fi products for interoperability.
- Conducts testing and awards the "Wi-Fi CERTIFIED" seal (e.g., Wi-Fi 6, Wi-Fi 7).

Standardization Process

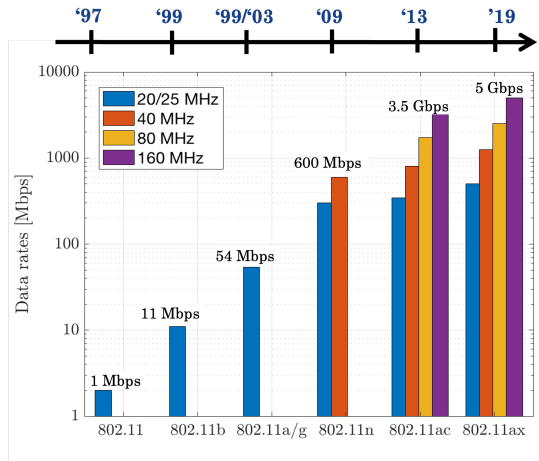


Evolution of Wi-Fi Standards

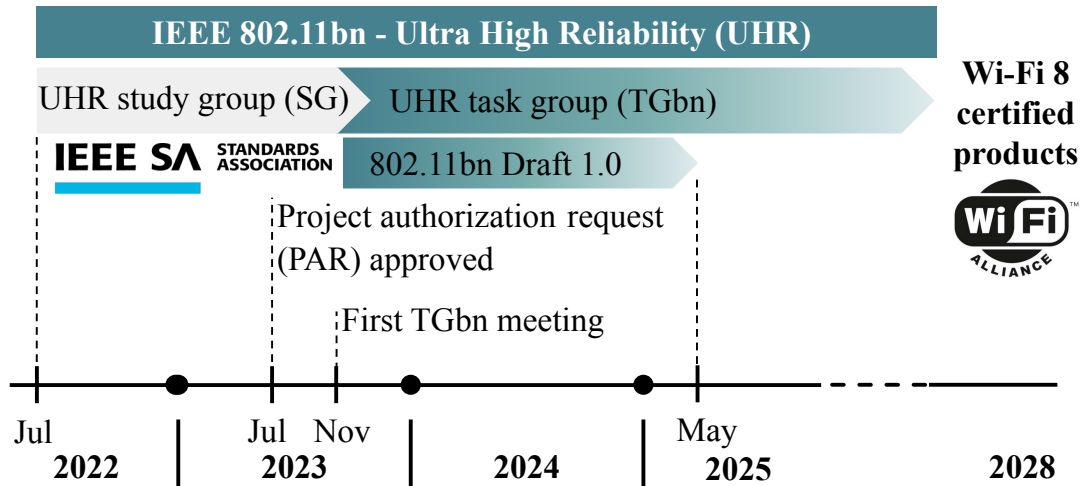
- **802.11b (1999):** First widely adopted standard, 11 Mbps (2.4 GHz).
- **802.11a (1999):** 54 Mbps (5 GHz), less widely adopted due to cost.
- **802.11g (2003):** 54 Mbps (2.4 GHz), backward compatible with 802.11b.
- **802.11n (Wi-Fi 4 - 2009):** Introduced MIMO, up to 600 Mbps (2.4/5 GHz).
- **802.11ac (Wi-Fi 5 - 2013):** Higher throughput, primarily 5 GHz, multi-user MIMO (MU-MIMO).
- **802.11ax (Wi-Fi 6/6E - 2019/2020):** OFDMA, Target Wake Time, better performance in dense environments (2.4/5/6 GHz).
- **802.11be (Wi-Fi 7 - 2024):** Extremely High Throughput (EHT), wider channels, more spatial streams.

What the 802.11 defines?

- 802.11n (Wi-Fi 4) [2.4 & 5 GHz]
 - Single-user MIMO
 - Packet aggregation
- 802.11ac (Wi-Fi 5) [5 GHz]
 - Multi-user MIMO (DL)
 - Channel bonding
- 802.11ax (Wi-Fi 6) [5 & 6 GHz]
 - OFDMA
 - Multi-user MIMO (UL)
- 802.11be (Wi-Fi 7) [2.4, 5 & 6 GHz]
 - Multi-link operation



IEEE 802.11bn (Wi-Fi 8)



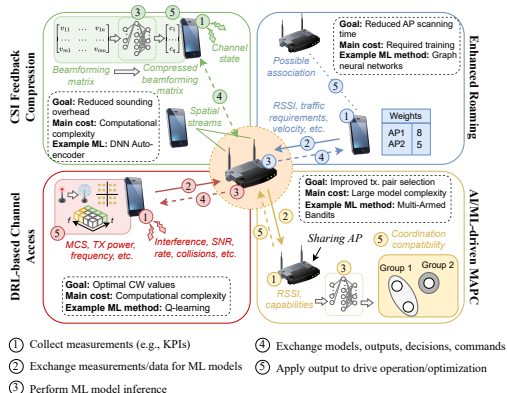
AI/ML in Wi-Fi standardization

- Progress:

- July 2022: The IEEE 802.11 establishes the AIML TIG
- March 2024: The TIG delivers a report on AI/ML for Wi-Fi use cases
- March 2024: The TIG becomes an IEEE 802.11 Standing Committee (SC)

- Use cases:

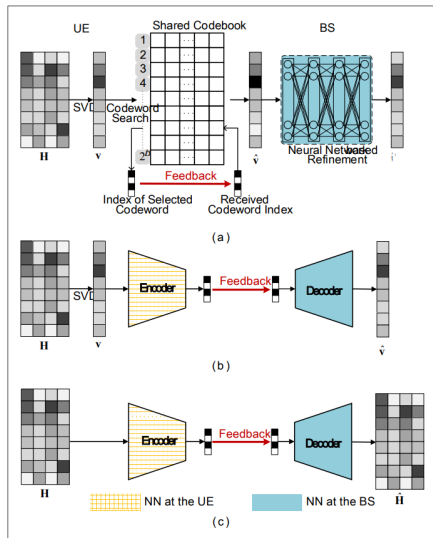
- CSI feedback compression based on Neural Networks
- AIML-based Roaming Enhancements
- DRL-based Channel Access
- ML-driven MAPC
- AIML model sharing



AIML-based CSI Feedback Compression

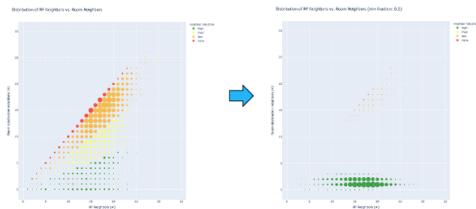
- Problem:
 - In 802.11ax/be the AP uses a sounding sequence for beamformee/beamformer communication
 - Sounding feedback airtime overhead increases with new features (e.g., MAP + many spatial streams)
- Benefit from ML:
 - CSI compression
 - Faster operation
 - Improved throughput

Guo, Jiajia, et al. "AI for CSI feedback enhancement in 5G-advanced." IEEE Wireless Communications (2022).



Enhanced Roaming

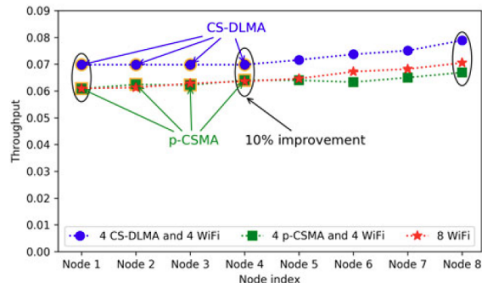
- Problem:
 - Roaming in dense environments can lead to very high roaming times
- Benefit from ML:
 - AI/ML is used to determine the probability of a client to roam to a specific AP, based on the learned client roaming patterns
 - Reduce the latency experienced by clients when they are operating in a highly roaming environment



Example: AIML is used to compute the 802.11k neighbor report list, where the number of APs returned to the STA is smaller. This allows STA to discover good neighbors for roaming quickly.

DRL-based Channel Access

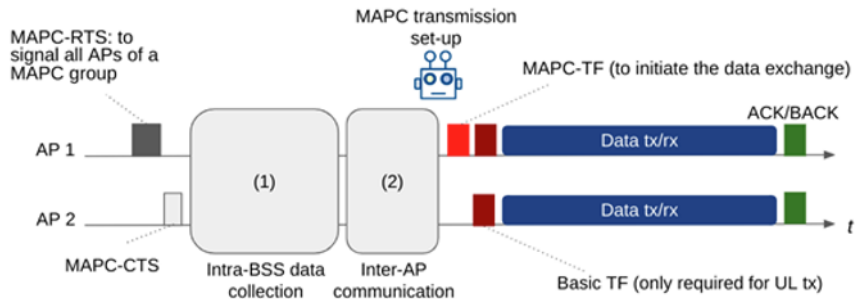
- Problem:
 - Current channel access methods based on CSMA/CA lack efficiency and fairness
- Benefit from ML:
 - Derive channel access policies whereby devices decide to transmit or no based on the environment
 - Enhance spectrum efficiency by reducing “empty” and “collision” slots



Y. Yu, S. C. Liew and T. Wang, "Non-Uniform Time-Step Deep Q-Network for Carrier-Sense Multiple Access in Heterogeneous Wireless Networks," in IEEE Transactions on Mobile Computing, vol. 20, no. 9, pp. 2848-2861, 1 Sept. 2021

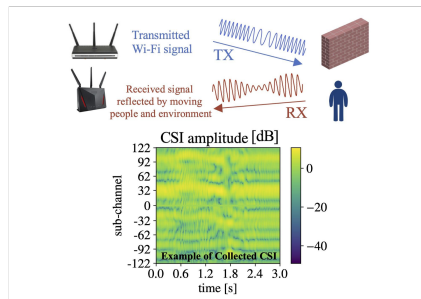
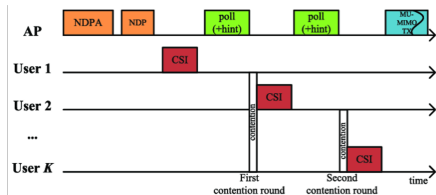
AIML-based Multi-AP Coordination

- Problem:
 - Dense multi-AP networks suffer from co-channel interference, multi-AP coordination (MAPC) becomes necessary
- Benefit from ML:
 - Selection of non-colliding groups → improved utilization of radio resources
 - Better informed decisions → fairness



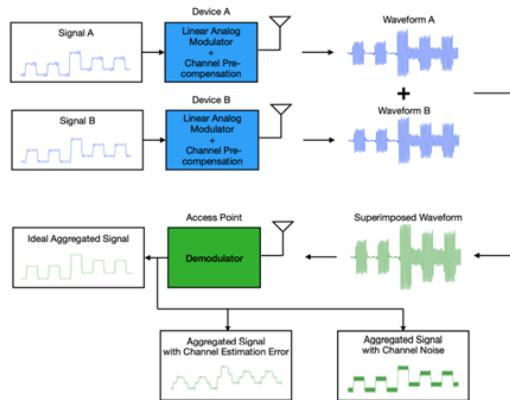
Wi-Fi sensing

- Problem:
 - Wi-Fi sensing leverages wireless signals to perceive and analyze environmental changes (e.g., human movement).
 - CSI provides detailed, fine-grained data about the path characteristics (amplitude and phase) of Wi-Fi signals as they travel from transmitter to receiver.
- Benefit from ML:
 - Interpret the complex Wi-Fi signals variations (reflection, absorption, diffraction) when affected by objects in their path.
 - Applications: Human activity recognition (fall detection, gesture/pose detection, breathing and heart rate monitoring, etc.), intrusion detection, through-wall sensing



Over-the-air-computation (OAC)

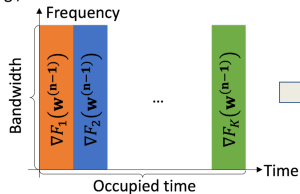
- OAC leverages interference from simultaneous transmissions for ML model aggregation/averaging
- The medium performs the computation (signal superposition)
- Reduce the number of computations
- The underlying communication infrastructure needs to be aware of the mechanism:
 - Schedule devices
 - Synchronization
 - Power control



Chen, M., Yang, Z., Saad, W., Yin, C., Poor, H. V., & Cui, S. (2020). A joint learning and communications framework for federated learning over wireless networks. *IEEE Transactions on Wireless Communications*, 20(1), 269-283

Over-the-air-computation (OAC)

e.g., with TDMA:



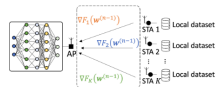
Frequency

Bandwidth

All STAs transmit simultaneously & gradient aggregation occurs in the wireless channel through signal superposition

Time

Occupied time

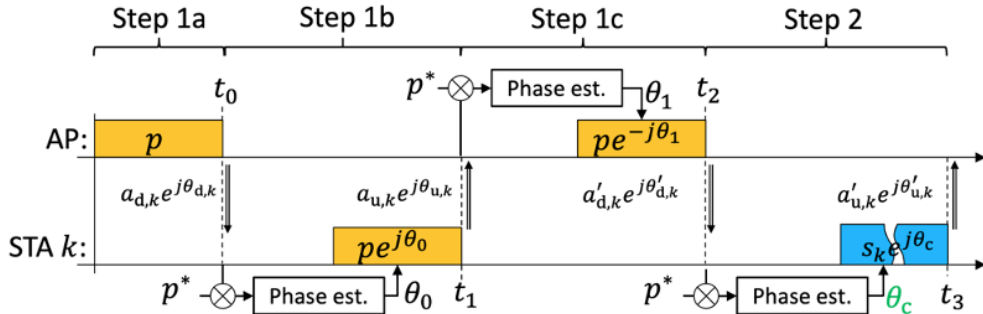


IEEE802.11-25/304r0, Feasibility Study of Phase-Synchronization for Wireless Federated Learning on WLAN (March, 2025)

Over-the-air-computation (OAC)

Challenge: The transmitted symbols (e.g., gradients) from all STAs should add up coherently at the AP side.¹

Solution: Add training sequences in the preamble and pilot signals in the data field to estimate CFO and remove it.



¹The Carrier Frequency Offset (CFO) represents the difference between the expected carrier frequency of the signal and the actual frequency received by the receiver (leads to inter-carrier interference (ICI)).

1. AI/ML in Wi-Fi standardization

2. Mock Test

Question #1

Which of the following frequency bands is commonly used by Wi-Fi devices?

1. 2.4 GHz
2. 900 MHz
3. 50 GHz
4. 1.8 GHz

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1. 2.4 GHz (Correct)
2. 900 MHz
3. 50 GHz
4. 1.8 GHz

Rationale: The 2.4 GHz band is commonly used for Wi-Fi, offering wider coverage but lower speeds than 5 GHz.

Question #2

Why can the 5 GHz band offer higher speeds than the 2.4 GHz band?

1. Because of the signal propagation characteristics of such a band
2. Because of the higher amount of bandwidth available in such a band
3. Because the Wi-Fi protocol is better for the 5 GHz band
4. Because $5\text{ GHz} > 2.4\text{ GHz}$

Answer #2

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(Correct)**
3. Because the Wi-Fi protocol is better for the 5 GHz band
4. Because $5\text{ GHz} > 2.4\text{ GHz}$

Rationale: The bigger bandwidth of the 5 GHz band allows for using wider channels (e.g., 80 MHz). Apart from that, since more spectrum is available, there can potentially be less interference among BSSs in the 5 GHz band.

Question #3

What does SSID stand for in the context of Wi-Fi?

1. A unique identifier for a wireless network
2. A type of Wi-Fi antenna
3. A security protocol for Wi-Fi
4. The maximum speed of a Wi-Fi network

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Rationale: An SSID (Service Set Identifier) is the name of a Wi-Fi network that users see and connect to.

Question #4

Which of the following Wi-Fi technologies allows for multiple users to transmit and receive data within the same frequency band using a single antenna?

1. Multiple-Input, Multiple-Output (MIMO)
2. Orthogonal Frequency-Division Multiple Access (OFDMA)
3. Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)
4. Spatial Multiplexing (SM)

Answer #4

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2. **Orthogonal Frequency-Division Multiple Access (OFDMA) (Correct)**
3. Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)
4. Spatial Multiplexing (SM)

Rationale: OFDMA is a feature that allows an AP to communicate with multiple clients simultaneously by dividing the channel into smaller sub-channels. MIMO and Spatial Multiplexing deal with multiple data streams to one or more devices, while CSMA/CA is the basic access method.

Question #5

What characteristic defines the “orthogonality” between subcarriers in an OFDM system?

1. They are separated by a frequency gap equal to the highest data rate.
2. The peak of each subcarrier spectrum aligns with the nulls of all other subcarrier spectra.
3. They always transmit data in exactly the same phase.
4. Their amplitudes are perfectly balanced across the entire bandwidth.

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Rationale: The orthogonality in OFDM means that even though the subcarriers are closely packed and overlap in the frequency domain, they can be separated at the receiver without interference. This is achieved by spacing the subcarriers such that the peak of one subcarrier's spectrum coincides with the zero-crossing (null) of all other subcarriers, allowing for efficient use of bandwidth without the need for guard bands between subcarriers.

Question #6

In the context of Multi-Armed Bandit problems, what fundamental trade-off must an algorithm balance to maximize its cumulative reward?

1. Bias-variance trade-off
2. Overfitting-underfitting trade-off
3. Exploration-exploitation trade-off
4. Supervised-unsupervised trade-off

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3. **Exploration-exploitation trade-off (Correct)**
4. Supervised-unsupervised trade-off

Rationale: Multi-Armed Bandit problems inherently involve the exploration-exploitation dilemma, whereby the agent decides whether to pull the arm that has yielded the best results so far (exploit), or try other arms with potentially higher rewards (explore).

Question #7

In the context of Reinforcement Learning, which of the following best describes a common strategy to dynamically adjust the exploration rate (ϵ) over time to optimize learning?

1. Keep ϵ constant at a high value to ensure maximum exploration.
2. Increase ϵ over time to progressively explore more.
3. Decrease ϵ over time to shift from exploration to exploitation.
4. Set ϵ to zero from the beginning to always exploit the best known action.

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Rationale: Starting with a relatively high ϵ encourages initial exploration, while gradually decreasing it allows exploiting the best actions as more expertise is gained over time.

Question #8

What is the primary issue caused by the “gradient vanishing” problem in deep neural networks?

1. The network learns too quickly, falling into local minima.
2. Gradients become too large, causing instability.
3. Gradients become too small, hindering effective weight updates in previous layers.
4. Gradients are not updated, thus the model cannot learn.

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Rationale: When using activation functions that saturate (like sigmoid or tanh), their derivatives become very small for large positive or negative inputs. As gradients are multiplied through many layers during backpropagation, these small values accumulate, causing the gradients for earlier layers to shrink exponentially and become effectively zero, preventing those layers from learning.

Question #9

The Upper Confidence Bound (UCB) algorithm for Multi-Armed Bandits aims to balance exploration and exploitation by selecting the arm that maximizes the sum of its empirical average reward and a confidence term. What does this confidence term primarily account for?

1. The amount of past reward received from that arm.
2. The current standard deviation of rewards for that arm.
3. The uncertainty or number of times that arm has been played.
4. The total number of plays across all arms.

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Rationale: The confidence term in UCB (e.g., proportional to $\sqrt{\frac{\ln N_t}{n_i}}$ where N_t is total plays and n_i is plays for arm i) explicitly accounts for the uncertainty in the estimate of an arm's mean reward. Arms that have been played fewer times have higher uncertainty (and thus a higher confidence bound), encouraging more exploration.

Question #10

In neural networks, what is the primary mechanism by which “Dropout” regularization helps to prevent overfitting?

1. It adds a penalty to the loss function based on the magnitude of the weights.
2. It randomly sets a fraction of neuron activations to zero during training, forcing the network to learn more robust features.
3. It stops the training process early when the validation error starts to increase.
4. It uses a smaller learning rate for weights connected to neurons with high activation.

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3. It stops the training process early when the validation error starts to increase.
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Rationale: Dropout prevents overfitting by randomly discarding a certain number of neurons and their connections during each training iteration. This forces the network to learn more redundant and robust features, as no single neuron can rely too heavily on the presence of another specific neuron.